

BRIDGE ENGINEERING CHALLENGE



Lesson Plan & Student Workbook

A Hands-On STEM Experience for Young Engineers

Grade Level 3 – 5	Duration 60 – 75 minutes	Subject STEM / EVS / Engineering	Difficulty Beginner ★■■■
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■ Meera's Big Problem — The Story Hook

Meera needs to visit her grandma who lives on the other side of a river. But there is no bridge! Today, **YOU** will become an engineer and help Meera by designing, building, and testing your very own bridge. Along the way, you will discover how real bridges in India work and what makes them so strong.

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1. Learning Objectives

By the end of this lesson, students will be able to:

- Define what a bridge is and explain its purpose in everyday life.
- Identify and label the three main parts of a bridge (deck, pillars/supports, span).
- Name at least two types of bridges (beam and arch) and give a real-world Indian example of each.
- Recall at least one famous Indian bridge and describe why it is significant.
- Apply the engineering design process: plan → build → test → improve.
- ⇒ Record scientific observations and reflect on how to improve their design.
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2. Key Vocabulary

Bridge	A structure built to cross water, roads, valleys, or other gaps safely.
Span	The horizontal distance covered from one support to another.
Gap	An open space or hole that a bridge must cross.
Pillar / Support	A vertical structure (column or pier) that holds the bridge up.
Deck	The flat surface of the bridge that people, vehicles, or trains use.
Load	The total weight a bridge must carry — its own weight plus everything on it.
Engineer	A person who uses science and math to design and build structures.
Arch	A curved shape that distributes weight outward and downward, making it very strong.

3. What Is a Bridge?

A **bridge** is a special structure — a kind of path — built to safely cross water, roads, valleys, railway tracks, or any other large gap. Bridges allow people, vehicles, and trains to travel without going around the obstacle, saving time and connecting communities.

■ Think About It!

Can you think of a bridge you have seen near your home or town? What was it crossing — a river, a road, or a railway track? Share your answer with a classmate!

4. Why Do We Need Bridges?

Bridges are essential infrastructure that improve quality of life in many ways:

■■ Connecting Communities

Bridges link villages, towns, and cities across rivers and valleys that would otherwise be impossible to cross. This is vital in a country like India with thousands of rivers.

■ Education Access

Bridges help children reach school safely during monsoon floods. Along rivers such as the Brahmaputra and Godavari, footbridges can mean the difference between a child attending school or missing months of education.

■ Reducing Traffic

City flyovers — elevated beam bridges — allow traffic to flow smoothly above busy intersections. Bengaluru, Chennai, and Delhi each have dozens of flyovers reducing congestion and travel times.

■ Island Connectivity

Railway bridges connect islands and remote regions to the mainland, enabling trade, tourism, and emergency services. The Pamban Bridge in Tamil Nadu is a celebrated example.

5. Bridges in India — Spotlight

■ Howrah Bridge Kolkata, West Bengal	One of the busiest cantilever bridges in the world, spanning the Hooghly River. Built in 1943, it carries an estimated 100,000 vehicles and 150,000 pedestrians every day. It is a proud symbol of Kolkata.
■ Pamban Bridge Rameswaram, Tamil Nadu	India's first sea bridge, built in 1914. It connects Rameswaram Island to mainland India across the Palk Strait. The bridge includes a bascule section that opens to allow ships to pass — an engineering marvel still in operation today.
■ Living Root Bridges Meghalaya, Northeast India	A remarkable example of bio-engineering! The Khasi and Jaintia tribal communities train the roots of rubber fig trees over rivers and streams. These living bridges grow stronger over time and can last hundreds of years.
■ Atal Setu (MTHL) Mumbai, Maharashtra	Opened in 2024, the Mumbai Trans Harbour Link is India's longest sea bridge at 21.8 km. It connects Mumbai to Navi Mumbai, holding thousands of cars and cutting travel time from over an hour to just 20 minutes.
■ ■ Bandra-Worli Sea Link Mumbai, Maharashtra	A cable-stayed bridge with a 5.6 km total length. Its tall pylons and hundreds of steel cables distribute the load of vehicles across the Arabian Sea inlet. It is an iconic part of the Mumbai skyline.

6. Types of Bridges

Beam Bridge

A beam bridge is the simplest type — essentially a flat horizontal beam resting on two or more supports at either end. The deck lies directly on the supports, and the weight of the load pushes down on the beam.

How it works: The beam bends slightly under load; the top is compressed (squished) and the bottom is in tension (stretched). Engineers choose strong materials like reinforced concrete or steel to resist these forces.

Indian Examples: Village wooden plank bridges, urban flyovers in Chennai and Delhi, railway bridges across flat plains.

Try It! Place a ruler between two equal stacks of books. That's a beam bridge! Put a small eraser in the center and watch it bend slightly.

Arch Bridge

An arch bridge uses a curved shape to gain its strength. Instead of the load pushing straight down, the arch redirects forces outward and downward into the ground at the base of each side. This is called compression.

How it works: The arch's curved shape means weight is distributed along the curve, not just at one point. This allows arch bridges to span much longer distances than simple beam bridges.

Indian Examples: Howrah Bridge (technically a cantilever truss but arch-inspired), old stone arch bridges in Rajasthan, Kerala, and Kashmir built during the Mughal era.

Try It! Make an arch with both hands by pressing your fingertips together above your head. Ask a friend to push down on the top — feel how strong it is!

7. Parts of a Bridge

■ ■ Deck	The flat surface of the bridge that people walk or vehicles drive on. It must be smooth, strong, and able to carry the full live load.
■ ■ Pillars / Supports / Piers	The vertical columns that hold the bridge deck up. They transfer the weight of the bridge down into the ground or river bed. Taller pillars allow ships to pass underneath.
↔ ■ Span	The distance the bridge stretches from one support to the next. Longer spans need more advanced engineering. The Bandra-Worli Sea Link has a main span of 250 metres!
■ Abutments	The structures at each end of the bridge that anchor it to the ground and absorb the horizontal forces pushing outward from the span.
■ Cables / Trusses (advanced bridges)	On suspension and cable-stayed bridges, steel cables transfer the deck's load up to tall towers. Trusses are triangular steel frameworks that add strength without too much extra weight.

8. What Makes a Bridge Strong?

A bridge must carry two kinds of load:

- **Dead Load:** The weight of the bridge structure itself — the deck, pillars, cables, etc.
- **Live Load:** The weight of everything on the bridge — vehicles, trains, pedestrians, and even wind!

Engineers use several strategies to make bridges strong enough to carry both loads safely:

- ✓ Choose strong materials such as reinforced concrete, structural steel, or naturally strong materials like tree roots.
- ✓ Use geometric shapes — especially **triangles** — which distribute forces evenly and resist bending.
- ✓ Add more pillars or supports to shorten each individual span and reduce bending stress on the deck.
- ✓ Use an arch or cable system to redirect downward forces outward and into the ground.
- ✓ Test the design thoroughly with computer simulations and physical load tests before opening.

■ Indian Spotlight: Atal Setu (Mumbai Trans Harbour Link)

At 21.8 km, Atal Setu is India's longest sea bridge. It rests on 5.6 km of pre-stressed concrete piles driven into the seabed. Engineers designed it to withstand high winds, sea salt corrosion, and the weight of thousands of vehicles every day. It opened in January 2024!

9. Build Activity — Design, Build & Test Your Bridge

■ Materials Needed

2 equal stacks of books (the "riverbanks") • Paper, cardboard, or popsicle sticks • Tape or clay (optional) • Coins for load testing • Pencil and this worksheet

STEP 1

Plan & Design ■

Draw your bridge design in the space below. Think like an engineer!

- How long does the bridge need to be to cross the gap?
- Where will the pillars (supports) go?
- What shape will the deck be?
- Will you use a beam or arch design — or your own invention?

STEP 2

Gather Materials ■

Collect your materials. Measure or estimate the gap between your two book stacks. Your bridge deck must span the full gap and rest on each stack on both sides.

STEP 3

Build ■■

Construct your bridge using your design as a guide.

- It's okay to make small changes as you build — real engineers do this too!
- Make sure the deck lies flat and is stable before testing.

STEP 4

Test with Coins ■

Place coins ONE BY ONE in the centre of your bridge deck.

- Count carefully — each coin adds to the load!
- Stop when the bridge bends noticeably or collapses.
- Record your result in the observation box on the next page.

STEP 5

Observe, Record & Improve ⇒■

Write or draw what happened:

- Did the bridge hold steady, bend slowly, or collapse suddenly?
- Where did the failure happen — the centre, the edges, or the supports?
- What would you change in your next design to hold more coins?

■ My Bridge Design Sketch

(Draw your bridge design here)

Observation Record

Bridge type I chose (beam / arch / other):	
Materials used:	
Number of coins held before failure:	
What happened to the bridge (circle one):	held steady / Bent slowly / Collapsed
Where did the failure occur?	
One thing I would change:	

10. Reflection Questions

Think carefully about what you learned and experienced today. Answer these questions in full sentences in your notebook or on the back of this sheet.

1. In your own words, what is a bridge? Why are bridges important for people?
2. Name one famous Indian bridge you learned about today. Where is it located and what makes it special?
3. What are the two types of bridges covered in this lesson? How are they different from each other?
4. Describe what happened when you loaded your bridge with coins. Was the result what you expected?
5. If you had extra materials (popsicle sticks, cardboard tubes, extra tape), what would you add or change to make your bridge hold more coins? Explain your reasoning.
6. Engineers always test and improve their designs. Can you think of another job or activity in life where "test and improve" is important? Explain.

11. Quick Quiz

Circle the best answer for each question. Good luck, Engineer! ■

Q1. A bridge helps us ____

- a) Eat food
- b) Cross a gap safely
- c) Sleep comfortably
- d) Grow plants

Q2. Howrah Bridge crosses the Hooghly River in ____

- a) Mumbai
- b) Chennai
- c) Kolkata
- d) Bengaluru

Q3. The flat part of a bridge that you walk or drive on is called the ____

- a) Pillar
- b) Span
- c) Deck
- d) Cable

Q4. An arch bridge is strong because ____

- a) It is very wide
- b) Its curved shape spreads forces outward
- c) It has no pillars
- d) It is made only of wood

Q5. Which of these is an example of a Living Root Bridge?

- a) Howrah Bridge, Kolkata
- b) Pamban Bridge, Tamil Nadu
- c) Meghalaya, Northeast India
- d) Atal Setu, Mumbai

Q6. The weight of the bridge structure itself is called ____

- a) Live Load
- b) Dead Load
- c) Arch Load
- d) Span Load

Q7. Atal Setu (Mumbai Trans Harbour Link) is notable because it is ____

- a) India's oldest bridge
- b) Made from tree roots
- c) India's longest sea bridge at 21.8 km

d) Located in Rajasthan

Answer Key

■ Answers

Q1: b) Cross a gap safely | Q2: c) Kolkata | Q3: c) Deck | Q4: b) Its curved shape spreads forces outward | Q5: c) Meghalaya, Northeast India | Q6: b) Dead Load | Q7: c) India's longest sea bridge at 21.8 km

12. Extension Ideas

■ Science Extension	Fold a single sheet of A4 paper into a corrugated (accordion/zigzag) shape and test how many coins it holds compared to a flat sheet. Why does folding increase strength?
■ Social Studies Extension	Research one bridge from another country (e.g., Golden Gate Bridge, USA; Tower Bridge, UK; Millau Viaduct, France). Compare it with Howrah Bridge — length, material, and purpose.
■ Art Extension	Draw or paint a detailed bridge of your own invention. Label all its parts using the vocabulary from this lesson.
■ Maths Extension	If your bridge held 10 coins and each coin weighs 5 grams, how much weight did it hold in total? What if you added 5 more coins — what would the new total be?
■ Language Extension	Write a short story (1–2 paragraphs) from Meera's perspective as she crosses her new bridge for the first time to visit her grandma. How does she feel?

■ ***Congratulations, Engineer!*** ■

You helped Meera cross her bridge. Keep building, keep improving!

Bridge Engineering Challenge Lesson Workbook • Designed for Grades 3–5 • STEM / EVS